

HEFT: Hierarchical Enhanced Fusion Transformer for RGB-D Salient Object Detection

Hanzhong Tan, Shuangbing Wen, Li Zhu, Haifeng Huang, Lingfeng Zhang, Jun Li and Tao Hu

Abstract—RGB-D Salient Object Detection (SOD) is widely used in robot navigation tasks, which helps robots identify and track salient objects in the environment to achieve efficient path planning and obstacle avoidance. However, traditional RGB-D SOD models based on encoder-decoder structures face limitations in extracting high-quality features from different modalities. To solve this dilemma, we propose a Hierarchical Enhanced Fusion Transformer (HEFT) model for RGB-D SOD. This model employs an encoder-converter-decoder architecture to improve the effective fusion and efficient extraction of multimodal information. The core component, the Cross-Modal Cross-Attention Enhanced Transformer Converter (C2E-TC), enables efficient cross-modal competitive learning. Additionally, we designed a Token-based Dual-task Dense Prediction (TDTDP) network, which captures both global and local features to enhance performance in dense prediction tasks. Experimental results demonstrate that HEFT achieves leading performance across several evaluation metrics on benchmark datasets, including NJUD, NLPR, and DUTLF-Depth, validating its effectiveness and superiority in RGB-D SOD tasks.

I. INTRODUCTION

By leveraging RGB-D Salient Object Detection (SOD), robot navigation systems can identify and track key objects within their environment, thereby enabling efficient path planning and obstacle avoidance [1]. As a critical technology, SOD empowers robots to distinguish salient objects or obstacles, allowing them to disregard extraneous information and concentrate on crucial targets [2]. This capability is essential for enhancing the accuracy, efficiency, and stability of tasks such as autonomous navigation [3] and obstacle avoidance [4]. By integrating SOD, target tracking technology significantly improves a robot's ability to efficiently track vital objects or dynamic entities efficiently, even in challenging environments, ultimately facilitating autonomous operations [5].

Hanzhong Tan, Shuangbing Wen, Li Zhu, Haifeng Huang, Jun Li and Tao Hu are with College of Intelligent Systems Science and Engineering, Hubei Minzu University, Enshi 445000, China (e-mail: 202330302@hbmzu.edu.cn; 202330267@hbmzu.edu.cn; 2009025@hbmzu.edu.cn; 202430317@hbmzu.edu.cn; lijun72213@163.com).

Tao Hu is corresponding author of this study (phone: 86 15549126665; e-mail: hutao_es@hbmzu.edu.cn).

Li Zhu and Tao Hu are with Hubei Engineering Research Center of Selenium Food Nutrition and Health Intelligent Technology, Enshi 445000, China

Lingfeng Zhang is with Department of Electrical Engineering and Information Systems, The University of Tokyo, Tokyo 113-8656, Japan (e-mail: zhanglf_scholar@outlook.com).

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Compared to traditional depth models that rely solely on RGB modality, which provide only two-dimensional color and texture information and lack perception of the physical and three-dimensional structure of the scene, utilizing RGB-D image pairs can significantly enhance SOD performance [6], particularly in challenging environments characterized by low foreground-background contrast or significant background clutter. The depth of information offers several key advantages. The integration of depth cues significantly improves model discrimination [7]. For instance, depth information allows the model to correctly infer spatial hierarchical relationships between object percepts using relative distances in scenarios with high color similarity [8]. In scenarios with complex or diverse background textures, depth information can distinctly separate the object from the background, enabling the model to further refine its recognition of salient objects by factoring in spatial position and distance [9]. In the multimodal input of RGB-D, RGB and depth information complement each other: when RGB features struggle to differentiate salient objects, the geometric and distance information provided by the depth map compensates for this limitation, facilitating a more robust and accurate detection [10].

For RGB-D SOD, many approaches usually perform multimodal feature fusion at the decoding stage, which is insufficient for extracting high-quality deep multimodal information features and accomplishing dense prediction of saliency maps. Our motivation is to design a dedicated converter for focusing on complementary feature fusion of different modalities in advance prior to decoding, to realize the competitive fusion of complementary features of different modalities, to complete the transition from encoding space to decoding space, and to leverage boundary features during decoding to refine the dense prediction of saliency maps. Specifically, the converter enhances feature extraction across attention units and improves the efficiency of self-modal competition during fusion, which is beneficial for the initial capture of important features during decoding, and the boundary information is utilized during decoding to further highlight salient object boundary features to accomplish dense prediction. This allows the robot to effectively focus on the most salient objects in the environment, thereby facilitating more precise path planning and collision avoidance.

We outline our principal contributions below:

- We propose Hierarchical Enhanced Fusion Transformer (HEFT), a novel method for RGB-D SOD that assist the robot in obstacle avoidance and navigation. It leverages an encoder-converter-decoder structure to realize the

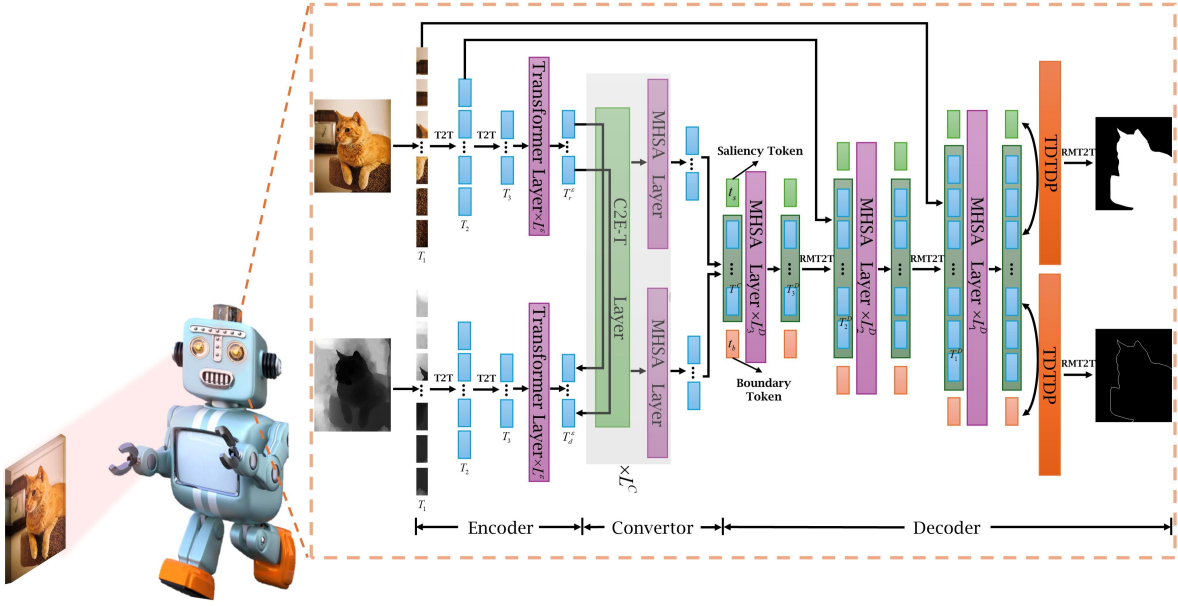


Fig. 1. Specific structure of the HEFT model. It uses a pre-trained encoder to extract multilevel tokens T_i at levels 1/4, 1/8 and 1/16. The converter then fuses the RGB and depth feature and transitions it to the decoding space. Finally, the decoder performs token-based dual-task dense prediction after up-sampling and completes decoding.

interactive expressiveness of different features.

- We formulate an enhanced cross-attention converter to improve cross-modal feature fusion. The salient features are extracted by competitive learning across different modalities. And the subsequent use of multiple stacked Transformer layers to obtain long-distance attention effectively. We also construct a token-based dual-task dense prediction (TDTDP) model for the RGB-D SOD dense prediction. While the TDTDP realizes the computation of global cross-attention and local cross-attention across token-style.
- In extensive evaluations on seven public RGB-D datasets (NJUD [11], NLPR [12], DUTLF-Depth [13], ReDWeb-S [14], STERE [15], SSD [16], SIP [17]), our model outperforms related works [18][19][20][21] and establishes new SOTA performance.

II. RELATED WORK

SOD finds widespread application in various computer and robotic vision tasks [1]. Early RGB-D SOD methods primarily relied on manually designed features. A representative example is the model by Feng et al. [22], which was built upon the concept of local background enclosure and utilized background cues to enhance detection accuracy. Similarly, Song et al. [23] employed multi-scale region segmentation, combining multi-level feature comparisons, weighting factors, and high-level positional priors for detection. Despite being effective in specific environments, these traditional approaches exhibit limited adaptability to complex and diverse scenes, which reduces their overall performance. The rise of deep learning has promoted the widespread adoption of Convolutional Neural Networks (CNNs) in SOD. CNN-based models can learn subtle features in the scene to significantly improve detection accuracy. For instance, A structure-aware dual pyramid network was proposed by Xu et al. [24] to

explicitly encode spatial location and covariance through a self-attention mechanism. The work [6] introduced CIR-Net, which employs an improved CNN architecture to further improve RGB-D SOD performance. But CNNs limit the model from exploring long-range features.

Recently, introducing self-attention mechanisms has led to significant advancements in computer vision tasks through models such as the Vision Transformer (ViT) [19]. Compared to CNNs, Vision Transformers possess a stronger ability for global information modeling, which has led to the emergence of many representative models based on ViT in the field of SOD [19][21]. Zheng et al. [25] introduced the Transformer-based Hierarchical Dynamic Decoder (THDDNet) for image SOD, using a dynamic dual upsampling approach to build a hierarchical dynamic decoder which integrates transformer features across different scales to achieve high-fidelity saliency detection. Moreover, the work by Wang et al. [26] centered on using a transformer decoder architecture for boosting information fusion and enhancement, termed IFFM and IFEM, in the context of RGB-D SOD. EM-Trans [27] is an edge-driven model that efficiently utilizes boundary information within a dual-frequency decomposition framework.

III. HIERARCHICAL ENHANCED FUSION TRANSFORMER

Fig. 1 shows the specific structure of the HEFT model. Its main components include a pre-trained transformer encoder, a cross-mode cross-attention enhanced transformer converter, and a dual-task transformer decoder.

A. Pre-training Transformer Encoder

We use the same pre-trained model as most RGB-D SOD task models as the encoder, and we use the T2T model [28] as the encoder for HEFT. The T2T model mainly contains re-structurization and soft split operations, and establishes

long range dependencies through the Self-attention (SA) Transformer layer.

$$T'_i = MLP(SA(T_i)) \quad (1)$$

In the re-structurization step, the tokens T_i are first transformed using the SA layer, and then obtain new tokens $T'_i \in \mathbb{R}^{l \times c}$. This process is Equation (1). The token T'_i will then reshape the image in the spatial dimension as follows:

$$I_i = Reshape(T'_i) \quad (2)$$

The re-structurization operation will be followed by a soft split to simulate local messages and reduce the token length. When implementing the soft split step, it is necessary to split the image into overlapping patches of size $m \times m$, with an overlap size of s and zero padding of m in order to avoid information loss. The obtained image patch $I_i \in \mathbb{R}^{h \times w \times c}$ is expanded into a tagging sequence $T_o \in \mathbb{R}^{l_o \times cm^2}$, the length of which is given in the Equation (3).

$$l_o = \lceil \frac{h + 2n - m}{m - s} + 1 \rceil \times \lceil \frac{w + 2n - m}{m - s} + 1 \rceil = h_o \times w_o \quad (3)$$

The tokens obtained here will be fed into the next T2T procedure.

B. Cross-modal Cross-Attention Enhancement Transformer (C2E-TC)

Previous RGB-D SOD networks often use an encoder-decoder structure, but this network structure leads to inefficiency in extracting useful features. We design the C2E-TC network to solve this problem, which is a cross-attention-enhanced Transformer model consisting of a C2E-T model with a Transformer layer (MHSA layer) that possesses multi-head self-attention. The C2E-T model is shown in Fig. 2.

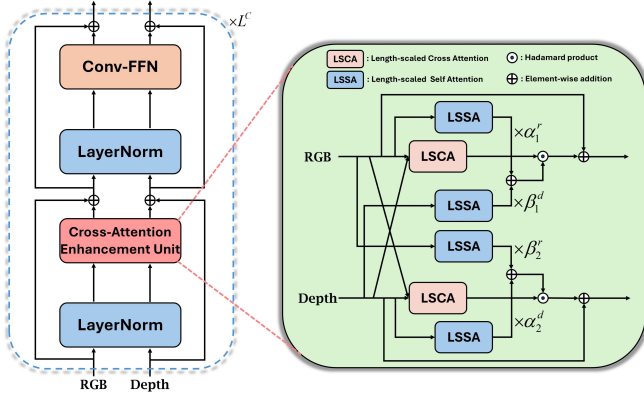


Fig. 2. The proposed Cross-modal Cross-Attention Enhancement Transformer (C2E-T) model is shown on the left and the Cross-Attention Enhancement Unit (CAEU) on the right.

RGB and Depth feature maps undergo cross-attention enhancement unit processing for efficient feature extraction. Firstly the layer normalized (LayerNorm) RGB and Depth feature maps are initially cross-modal cross-fertilized by our designed length-scaled Cross Attention (LSCA), and it can be described as Equation 4.

$$\begin{aligned} LSCA_r(Q_r, K_d, V_d) &= Softmax(\frac{\tau_r \log n}{\sqrt{d}} Q_r K_d^T) V_d \\ LSCA_d(Q_d, K_r, V_r) &= Softmax(\frac{\tau_d \log n}{\sqrt{d}} Q_d K_r^T) V_r \end{aligned} \quad (4)$$

where, the RGB feature maps are obtained as Q_r , K_r , and V_r by three linear mappings, and Depth feature maps are obtained as Q_d , K_d , and V_d by the same operation. Recent study [29] have found that output attention credibility decreases as the input sequence length increases in the attention mechanism. Therefore the scale factor of the attention mechanism is related to the length of the input sequence [29], and the test found that the scale factor multiplied by $\log n$ is beneficial to generalize the length. We introduce a learnable scale factor tensor τ on this basis to obtain better sequence length extrapolation performance.

Length-scaled Self Attention (LSSA) also employs a learnable length-scaled factor for better extrapolation performance. The query, key, and value vectors are derived by applying linear projections to the RGB feature maps, following which the attention weights are computed according to Equation 5.

$$\begin{aligned} LSSA_r(Q_r, K_r, V_r) &= Softmax(\frac{\tau_r \log n}{\sqrt{d}} Q_r K_r^T) V_r \\ LSSA_d(Q_d, K_d, V_d) &= Softmax(\frac{\tau_d \log n}{\sqrt{d}} Q_d K_d^T) V_d \end{aligned} \quad (5)$$

Where $LSSA_r$ and $LSSA_d$ are the computed obtained RGB feature maps attention, $LSSA_r$ will be used to multiply with the RGB feature maps after $LSCA_r$ attention computation and $LSSA_d$ will be multiplied with the Depth feature maps after $LSCA_d$ attention computation. Similarly, we can get the result $LSSA_d^d$ and $LSSA_d^r$ of the attention calculation of the Depth feature map twice as shown in Equation 6.

$$\begin{aligned} LSSA_d^d(Q_d, K_d, V_d) &= Softmax(\frac{\tau_d \log n}{\sqrt{d}} Q_d K_d^T) V_d \\ LSSA_d^r(Q_r, K_r, V_r) &= Softmax(\frac{\tau_r \log n}{\sqrt{d}} Q_r K_r^T) V_r \end{aligned} \quad (6)$$

The introduction of learnable coefficient tensor α_1^r , β_1^d , α_2^d , β_2^r maintains the healthy competition from the effective features of different modalities, and maintains the relative equilibrium between the two in different modalities. In addition, the results after adding T_r^ε and T_d^ε LayerNorm (LN) to maintain the feature diversity of different modalities. The output of the Cross-Attention Enhancement Unit (CAEU) can be expressed as the Equation 7.

$$\begin{aligned} CAEU_r^\varepsilon &= LSCA_r(\alpha_1^r LSSA_r + \beta_1^d LSSA_d^r) + LN(T_r^\varepsilon) \\ CAEU_d^\varepsilon &= LSCA_d(\alpha_2^d LSSA_d + \beta_2^r LSSA_r^d) + LN(T_d^\varepsilon) \end{aligned} \quad (7)$$

The Transformer model lacks attention to the local environment, and the convolutional operation is used in the C2E-T model to tune the feed forward network (Conv-FFN), which is designed to improve the lack of attention to the local area. After using the C2E-T multi-attention model, C2E-TC uses the MHSA layer on each RGB and depth sequence to embed the enhancement.

C. Dual-task Transformer Decoder

We use the up-sampling method of multilevel token fusion to implement up-sampling and token-based Dual-tasking with dense prediction to accomplish RGB-D SOD decoding.

1) *Up-sampling on multi-level fused tokens*: We sample patch tokens using the reverse T2T module with multi-head attention (RMT2T). The input patch tokens are projected, as with soft splits in T2T, by treating each token as if there were s overlapping $m \times m$ image patches in neighboring patches, and inversely using the Equation (3) to compute the image size, which is then restored to the original image size using the three times RMT2T. Accurate local feature structure information can be provided using multi-level feature fusion. The splicing and linear projection is used to merge T_2 and T_1 step by step, and the obtained up-sampled patch tokens are inputted to the MHSA layer to get the decoder tokens T_i^D at each layer i . The processing is shown in Equation 8.

$$T_i^D = MLP(MHSA(Linear([RMT2T(T_{i+1}^D), T_i]))) \quad (8)$$

Where $[.,.]$ denotes the connection from the embedding dimension of the token. MLP stands for multilayer perceptron.

2) *Token-based Dual-task Dense Prediction (TDTDP)*: Fig. 3 shows the proposed token-based dual-task dense prediction (TDTDP) model. TDTDP can aggregate task-relevant features and perform RGB-D SOD prediction, and the introduction of the boundary detection task into the SOD task can facilitate the exchange of object-boundary information to enhance prediction performance.

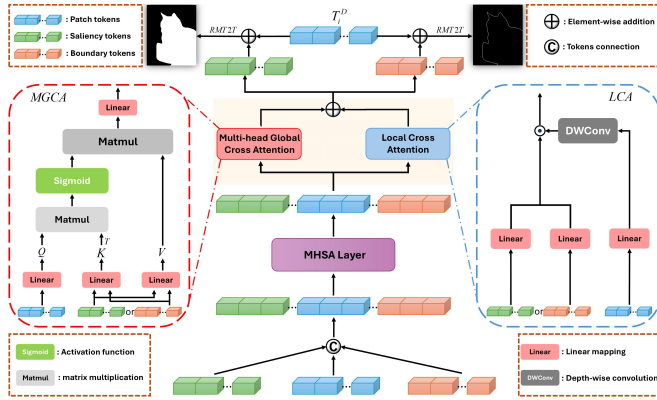


Fig. 3. Token-based Dual-task Dense Prediction (TDTDP) model, which maintains attention to global and local features of saliency and boundary tokens.

Therefore, we designed saliency token t^s and boundary token t^b , connected them with a sequence of patch tokens T_i^D , processed them using the L_i^D MHSA layer, and then inputted them to the MLP, as shown in Equation 9.

$$[t_i^s, T_i^D, t_i^b] = MLP(MHSA([t_{i+1}^s, T_i^D, t_{i+1}^b])) \quad (9)$$

where the two task tokens entered in each layer use the results of the previous layer, and the two task tokens have interacted with the patch tokens using multi-head self-attention, and the relevant task feature information is aggregated.

We perform dense prediction for both task-related tokens to produce saliency and boundary detection. The traditional attention mechanism lacks the extraction of local information, so multi-head global cross-attention (MGCA) and local cross-attention (LCA) is designed to maintain the extraction

of coarse-grained and fine-grained information for dual-task dense prediction. For saliency and boundary detection MGCA is specified by embedding T_i^D into the queries Q_{si}^D and Q_{bi}^D , embedding t_i^s and t_i^b into the keys K_{si} and K_{bi} , respectively, and then embedding t_i^s and t_i^b into the values V_{si} and V_{bi} . Learnable τ_i^s and τ_i^b are still used to maintain the length extrapolation performance. The equation for MGCA is given in Equation 10, where $t \in \{s, b\}$.

$$MGCA_{ti}^D = Sigmoid(\frac{\tau_i^t \log n}{\sqrt{d}} Q_{ti}^D K_{ti}^T) V_{ti} \quad (10)$$

The LCA model captures more local details by performing depth-wise convolution with a large kernel on patch tokens T_i^D and using the generative features as attention weights to reweight the saliency token t_i^s with the boundary token t_i^b using the Handmaid product as shown in Equation 11.

$$LCA_{ti}^D = DW C_{k \times k}^{ti} (Linear(T_i^D)) \odot Linear(t_i^t) \quad (11)$$

where $t \in \{s, b\}$, $DW C_{k \times k}^{ti}$ is a depth-wise convolution and $\odot$ is an elemental multiplication.

The obtained global attention features are summed with the local attention features to get the hybrid attention features, and finally, T_i^D is added to maintain the feature diversity as in Equation 12, where $t \in \{s, b\}$.

$$T_{ti}^D = MGCA_{ti}^D + LCA_{ti}^D + T_i^D \quad (12)$$

The saliency and boundary feature maps for different strata can be obtained after RMT2T up-sampling.

IV. EXPERIMENTS

A. Experimental Setup

1) *Datasets*: We validated the HEFT model on seven publicly available RGB-D datasets, NJUD [11] (1,985 image pairs), NLPR [12] (1,000 image pairs), DUTLF-Depth [13] (1,200 image pairs), ReDWeb-S [14] (3,179 image pairs), STERE [15] (1,000 image pairs), SSD [16] (80 image pairs), and SIP [17] (929 image pairs). The HEFT model was validated on seven publicly RGB-D SOD datasets. In order to fairly compare the previous works [18][19][20][21], the dataset was split with 700 NLPR, 1,485 NJUD, and 800 DUTLF-Depth images for training, and the remaining images for testing.

2) *Evaluation Metrics*: The evaluation of the HEFT model was conducted using four performance metrics. The model was evaluated using four standard metrics. The Maximum F-measure (maxF) [30] assesses precision-recall balance at an optimal threshold. The Maximum Enhanced-Alignment Measure (E_{max}^E) [31] integrates both pixel-wise and global-level statistical errors. Structure-measure (S_m) [32] quantifies region-wise and object-level structural similarities. Finally, the Mean Absolute Error (MAE) [30] computes the mean per-pixel absolute difference.

3) *Experimental Environment*: We conducted all experiments on a single NVIDIA TITAN RTX GPU, using the Pytorch framework and on CUDA version 10.2.

TABLE I

DETECTION PERFORMANCE COMPARISONS WITH SOTA ON SEVEN BENCHMARKS. OPTIMAL VALUES ARE HIGHLIGHTED USING BOLD TYPE.

Methods	Year	NUJUD				NLPR				DUTLF-Depth				ReDWeb-S				SSD				SIP				STERE			
		$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$
DFM-Net	2021	-	-	-	-	0.9230	0.9058	0.9552	0.0251	-	-	-	-	0.7054	0.6847	0.7824	0.1346	-	-	-	-	0.8831	0.8858	0.9245	0.0492	0.8980	0.8914	0.9393	0.0432
VST	2021	0.9241	0.9227	0.9546	0.034	0.9315	0.9204	0.9627	0.0237	0.9396	0.9444	0.9649	0.0263	0.7580	0.7596	0.8284	0.1114	0.8835	0.8659	0.9303	0.0446	0.9031	0.9153	0.9446	0.0399	0.9097	0.9028	0.9478	0.0396
SPSN	2022	0.9191	0.9173	0.9479	0.0318	0.9172	0.8959	0.9485	0.0256	0.9258	0.9249	0.9527	0.0289	0.7053	0.6923	0.7693	0.1353	0.8578	0.8384	0.8968	0.0511	0.8878	0.8943	0.9292	0.0445	0.9066	0.8990	0.9426	0.0361
CAVER-R50	2023	0.9204	0.9223	0.9511	0.0305	0.9291	0.9196	0.9619	0.0215	0.9314	0.9374	0.9605	0.0273	0.7299	0.7232	0.8003	0.1200	0.8740	0.8525	0.9164	0.0427	0.8936	0.9052	0.9327	0.0416	0.9137	0.9096	0.9502	0.0320
HEFT		0.9256	0.9249	0.9567	0.0327	0.9325	0.9218	0.9638	0.0236	0.9430	0.9468	0.9682	0.0244	0.7603	0.7678	0.8324	0.1099	0.8860	0.8686	0.9351	0.0409	0.9083	0.9175	0.9487	0.0368	0.9158	0.9102	0.9523	0.0364

B. Comparison With SOTA Methods

HEFT model is extensively compared with four SOTA methods on seven publicly available datasets. These SOTA models include DFM-Net [18], VST [19], SPSN [20], and CAVER-R50 [21]. The saliency maps for these benchmark methods were obtained either by running the authors' original code with default settings or by utilizing the maps directly supplied in their papers.

1) *Quantitative Analysis*: A comparative analysis of HEFT against other RGB-D SOD networks across the seven datasets is presented in TABLE I. For metrics S_m , $\max F$ and E_{max}^ζ , higher values indicate better performance, whereas for MAE, a lower value is preferred. As presented in Table I, the test results of HEFT on seven publicly available datasets are significantly better than the existing SOTA methods, resulting in a significant improvement in robot navigation performance. Fig.4 PR curve shows that HEFT has the best overall performance.

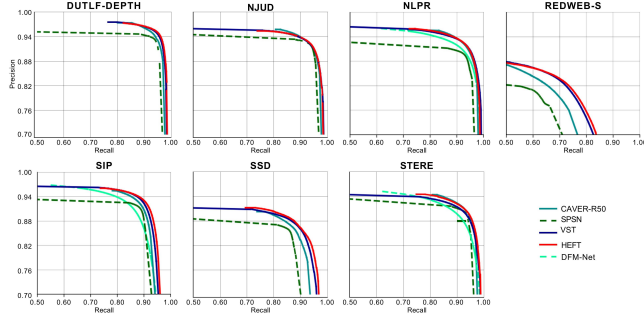


Fig. 4. Precision-Recall curves comparison with SOTA on seven datasets.

2) *Qualitative Analysis*: We selected the comparison results between indoor and outdoor environments, shown in Fig. 5. In Fig. 5, there is a wide variety of subject types, including objects affected by light and refractions (Column 1 and 2), detection of salient multi-objects (Column 5), objects with unclear and complex boundaries (Column 4-6, and 8), and cases featuring cluttered backgrounds (Columns 1, 6-9).

Our results unequivocally demonstrate the robustness and accuracy of the HEFT model in detecting salient objects in challenging environments. The model's high accuracy in detecting big salient objects, complex backgrounds, and its effective fusion of RGB and depth information with salient objects, which can be very beneficial for robots to navigate.

C. Ablation Studies

TABLE II shows that C2E-TC and TDTDP models achieve the best results in the ablation experiments. We conducted



Fig. 5. Example of image comparison of HEFT in ReDWeb-S dataset with SOTA methods.

TABLE II

ABLATION STUDY FOR CORE COMPONENTS WITH OPTIMAL VALUES IN BOLD.

models	DUTLF-Depth				STERE			
	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$	$S_m \uparrow$	$\max F \uparrow$	$E_{max}^\zeta \uparrow$	$MAE \downarrow$
Ec+Dc(RMT2T)	0.9395	0.9439	0.9665	0.0255	0.9084	0.9008	0.9466	0.0394
Ec+Dc(RMT2T+MHSA)	0.9401	0.9444	0.9654	0.0266	0.9121	0.9048	0.9474	0.0393
Ec+Dc(RMT2T+MHSA+TDTDP)	0.9402	0.9454	0.9656	0.0253	0.9156	0.9084	0.9505	0.0366
Ec+C2E-TC+Dc(RMT2T+MHSA+TDTDP)	0.9430	0.9468	0.9682	0.0244	0.9158	0.9102	0.9523	0.0364

experiments using the T2T encoder (Ec) and upsampling-based RMT2T to verify the performance of MHSA in the decoder (Dc), TDTDP, and C2E-TC. The experiments demonstrate that C2E-TC promotes cross-modal feature fusion efficiency, as well as global and local cross-attention perceptual performance of TDTDP across token styles.

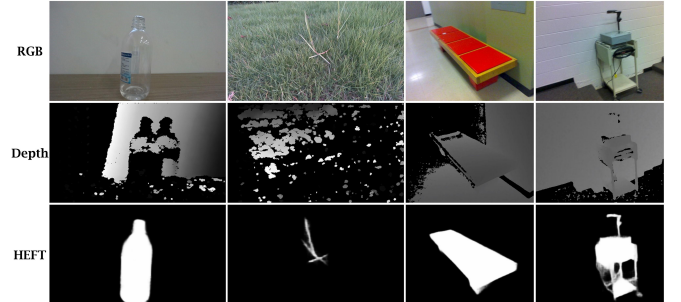


Fig. 6. Vision test results of Intel RealSense depth camera.

D. RealSense Camera Test Validation

The detection capability of HEFT was evaluated using depth images captured in indoor and outdoor environments with an Intel RealSense camera. The corresponding test

results are presented in Fig. 6. In Fig. 6, objects in the outdoor scene (Column 2) and in the indoor scene (Column 1, 3 and 4) are included, including objects with solid interference in the background (Column 2), including objects with severe interference in the detection environment (Column 2), including objects with interference from reflected light (Column 1 and 3), and including objects with the similar or same transparency as the background color (Column 1 and 4). As the test results in Fig. 6 show, the HEFT model still achieves excellent results in challenging environments.

V. CONCLUSION

The proposed HEFT is a hierarchical enhanced fusion transformer for RGB-D SOD. HEFT accomplishes cross-competition feature fusion using a Cross-modal Cross-attention Enhanced Transformer Converter (C2E-TC) for initial capture of important features prior to decoding. HEFT introduces a Token-based Dual-task Dense Prediction (TDTDP) network, focusing on both global and local feature extraction and processing, thereby improving navigation accuracy. A series of rigorous experiments conducted on seven benchmark datasets show that HEFT consistently outperforms other methods according to various metrics.

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